B

Biochemistry Cume

December 7, 2018

Total points: 100 points

1. A. Define each of the following,

a. Antiporters **b**. Symporters **c**. Uniporters **15 (5 + 5 + 5) points**

B. Calculate the free energy of transport of Ca^{2+} ions out of a cell, under physiological conditions, when the concentration of Ca^{2+} in the cytoplasm and the extracellular compartment are 0.5 μM and 2 mM, respectively. The membrane potential of the cell is – 60 mV. [Universal Gas constant = 8.315 $J.mol.^{-1}.K^{-1}$ and the Faraday constant = 96. 3 $kJ.V^{-1}$). **(25 points)**

2. **A.** What is/are the difference between ligand-gated and voltage-gated channels. Give examples of each class of channels. **10 (5 + 5) points**

B. Provide a cartoon representation of a nerve Action Potential and indicate the different phases that constitute a nerve action potential. (**20 points**)

C. Why do only a small number of sodium ions need to flow through the Na^+ channel to change the membrane potential significantly (**10 points**).

D. Assume that a channel obeys the concerted allosteric model Monod-Wyman-Changeux. The binding of ligand to the Relaxed (R) State (the open form) is 20 times as tight to the Tight (T) State (the closed form). In the absence of ligand, the ratio of closed to open channels is 10^5. If the channel is a tetramer, what is the fraction of open channels 1, 2, 3, and 4 ligands are bound. (***To receive credit, appropriate explanation/rationale should be provided).*** **(20 points)**